Tianhu Peng

PhD Researcher in Robotics

University College London (UCL) | London, United Kingdom

pengtianhu@gmail.com | Personal Website | Google Scholar | GitHub | LinkedIn

Research Profile

Robotics researcher specialising in **humanoid locomotion, whole-body motion generation, and embodied intelligence**. My research combines **reinforcement learning, imitation learning, predictive control, and model-based whole-body control** to develop robust controllers that transfer from simulation to physical robots. Experience spans academic research, industrial robotics at Tencent Robotics X, real-robot deployment on Unitree humanoids, and teaching in legged robotic systems.

Research Interests

Humanoid robotics · legged locomotion · whole-body motion generation · whole-body control · reinforcement and imitation learning · diffusion and flow-based motion policies · audio-conditioned motion generation · optimal control · predictive perception · sim-to-real transfer · embodied AI

Education

University College London (UCL)

Nov 2024 – Present

PhD in Robotics

London, UK

- Research on learning-based humanoid locomotion, whole-body motion generation, predictive perception, and robust sim-to-real deployment using the Unitree G1, Isaac Gym, Isaac Lab, and MuJoCo.
- Continued doctoral research after completing the first-year transfer viva at the University of Leeds.

University of Leeds

May 2023 – Oct 2024

PhD in Robotics

Leeds, UK

- Research on bipedal locomotion, adversarial motion priors, teacher–student learning, and vision-based locomotion; completed first-year doctoral training and transfer viva before moving to UCL.

Imperial College London

Oct 2020 – Oct 2021

MSc Human and Biological Robotics – Merit

London, UK

University of Sheffield

Sep 2017 – Jul 2020

BEng Mechatronics and Robotics Engineering – First Class Honours

Sheffield, UK

Research Experience

Tencent Robotics X Laboratory

Jul 2022 – Apr 2023

Robotics Researcher, Control Center Department

Shenzhen, China

- Designed and implemented model-based whole-body controllers for bipedal robots, including task-space tracking, contact constraints, force distribution, and hierarchical task prioritisation.
- Developed locomotion capabilities for flat and uneven terrain, push recovery, stair climbing, and coordinated whole-body motion.
- Integrated algorithms with physical robot hardware and contributed to real-time validation and debugging.

Teaching Experience

University College London

2025 – Present

Teaching Assistant – Legged Robot Systems

London, UK

- Support practical teaching in locomotion control, simulation, reinforcement learning, and robot software integration.
- Guide students through technical debugging, coursework, and laboratory demonstrations.

Imperial College London

Oct 2021 – Dec 2021

Graduate Teaching Assistant – Robotics

London, UK

- Assisted students with robotics course content, programming exercises, technical problem solving, and laboratory work.

Publications

[1] **T. Peng**, Y. Wang, L. Bao, Y. Jia, and C. Zhou.
AHEADance: Anticipatory Humanoid Dance from Streaming Audio.
Submitted to IEEE Transactions on Robot Learning (T-RL), 2026; arXiv:2611.13370. Project page.

[2] L. Bao, H. Zhu, Y. Pan, J. Wang, J. Yu, **T. Peng**, Y. Jia, D. Kanoulas, and C. Zhou.
Beyond Kinematic Tracking: Preserving Expressive Characteristics in Humanoid Whole-Body Control.
arXiv preprint arXiv:2610.19393, 2026. Under review.

[3] B. Long, L. Bao, **T. Peng**, J. Sun, and C. Zhou.
Reliability-Aware Sparse Route Memory for Round-Trip Vision-Language Navigation.
arXiv preprint arXiv:2609.34163, 2026.

[4] Y. Pan, L. Bao, **T. Peng**, and C. Zhou.
PAMoR: Parameterized Affective Motion Generation in Real Time for Humanoid Robots.
arXiv preprint arXiv:2608.28213, 2026. Under review at IEEE Robotics and Automation Letters.

[5] Z. Sun, L. Bao, **T. Peng**, J. Sun, and C. Zhou.
A High-Fidelity Digital Twin for Robotic Manipulation Based on 3D Gaussian Splatting.
Robot Learning, article 20260013, 2026.

[6] M. Ziegltrum, J. Jiao, **T. Peng**, C. Zhou, and D. Kanoulas.
Quadruped Parkour Learning: Sparsely Gated Mixture of Experts with Visual Input.
arXiv preprint arXiv:2604.19344, 2026.

[7] M. Carroll, **T. Peng**, L. Bao, C. Zhou, and Z. Li.
SCDP: Learning Humanoid Locomotion from Partial Observations via Mixed-Observation Distillation.
arXiv preprint arXiv:2603.09574, 2026.

[8] L. Bao, J. Humphreys, **T. Peng**, and C. Zhou.
Deep Reinforcement Learning for Robotic Bipedal Locomotion: A Brief Survey.
Artificial Intelligence Review, vol. 59, article 38, 2026.

[9] L. Bao, **T. Peng**, and C. Zhou.
Sim-to-Real Transfer in Deep Reinforcement Learning for Bipedal Locomotion.
Book chapter preprint, arXiv:2511.06465, 2025.

[10] **T. Peng**, L. Bao, and C. Zhou.
Gait-Conditioned Reinforcement Learning with Multi-Phase Curriculum for Humanoid Locomotion.
IEEE-RAS 24th International Conference on Humanoid Robots (Humanoids), pp. 1087–1093, 2025.

[11] L. Bao, Y. Pan, **T. Peng**, D. Kanoulas, and C. Zhou.
Hierarchical Intention-Aware Expressive Motion Generation for Humanoid Robots.
arXiv preprint arXiv:2506.01563, 2025.

[12] M. Song, M. Zhang, A. Lyu, L. Bao, **T. Peng**, and C. Zhou.
OmniDexter: A Modular Tendon-Driven Robotic Wrist with Enhanced Precision and Versatility.
7th UK Robot Manipulation Workshop, 2025.

[13] **T. Peng**, L. Bao, J. Humphreys, A. M. Delfaki, D. Kanoulas, and C. Zhou.
Learning Bipedal Walking on a Quadruped Robot via Adversarial Motion Priors.
Towards Autonomous Robotic Systems (TAROS 2024), LNCS 15052, pp. 118–129, 2024. **Best Paper Award**.

Complete and current list: Google Scholar profile.

Selected Research Projects

AHEADance: Streaming Humanoid Dance — Unitree G1

- Developed a robot-native streaming dance generator conditioned on heard audio and reliability-weighted forecasts of upcoming music (ListenAhead).
- Trained continuation from committed motion chunks with Commit Forcing and deployed the generated motion on a physical Unitree G1 through whole-body tracking.

Unified Humanoid Locomotion — Unitree G1

- Developed a recurrent, gait-conditioned policy for standing, walking, running, and gait transitions using curriculum learning and reward routing.
- Demonstrated stable standing, walking, and transition behaviours on a physical Unitree G1 humanoid robot.

Bipedal Locomotion with Adversarial Motion Priors — Unitree A1

- Combined adversarial motion priors with teacher–student reinforcement learning for bipedal walking across flat and challenging terrain.
- The resulting work received the TAROS Best Paper Award.

Predictive Perception and Vision-Fault Robustness — Humanoid locomotion

- Investigating predictive sensory-latent belief, reliability-aware fallback, memory provenance, and task-dependent sensory refresh under visual corruption and occlusion.

Whole-Body Motion and Expressive Humanoids — Motion generation

- Research on intention-aware expressive motion, whole-body tracking, latent motion representations, and continuous motion generation for humanoid robots.

Awards, Talks, and System Challenges

- **Best Paper Award**, Towards Autonomous Robotic Systems (TAROS), 2024.
- Invited presentation at NVIDIA Robotics Day on distillation-based locomotion training in Isaac Lab.
- Participated in the ATEC Humanoid Robot Track using a full-scale Unitree G1 for dynamic balance, outdoor traversal, and whole-body manipulation tasks.
- Best Poster recognition for work on learning bipedal walking on a quadruped robot.

Technical Skills

Robotics and Control	Humanoid locomotion, legged robots, whole-body control, optimal control, contact dynamics, motion tracking, and sim-to-real deployment
Learning Methods	Reinforcement learning, PPO, imitation learning, adversarial motion priors, diffusion policies, flow-based generation, curriculum learning, and teacher–student distillation
Programming	Python, C++, PyTorch, ROS, ROS2, DDS, Git, and Linux
Simulation and Tools	Isaac Gym, Isaac Lab, MuJoCo, PyBullet, Pinocchio, Unitree SDK, Docker, and OpenCV

Languages

Chinese – Native

English – Fluent

Last updated: 11 November 2026